

The SZL Governed-AI Estate

Live Verifiable Inference with Signed Receipts,
Machine-Checked Invariants, and Honest Trust Bounds

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THE HONEST HOOK

A live, 63-surface governed-AI estate where every inference emits a signed receipt, a handful of core invariants are machine-checked in Lean, and the advisory trust score is honestly capped below 1.0 — never claimed as proof it cannot deliver.

ABSTRACT

We present the SZL Governed-AI Estate: a running, publicly inspectable system for verifiable governed inference. The estate exposes 63 live holographic surfaces^{2,3} at a-11-oy.com/holographic, a living-anatomy telemetry layer that reports MEASURED GPU energy in joules whenever a hardware exporter is reachable (and honestly reports UNAVAILABLE otherwise, never a fabricated number)⁴, and a governed-inference receipt primitive by which each inference emits a signed, offline-verifiable attestation of what ran, with which configuration, and its hash. Its formal core is a set of exactly eight invariants — the locked-8, {F1,F4,F7,F11,F12,F18,F19,F22} at Lean kernel c7c0ba17 — machine-checked in Lean 4. Crucially, the estate's advisory trust functional Λ is published as Conjecture 1, explicitly not a theorem, its trust score is capped at 0.97 and never at 1.0, and the Byzantine-fault-tolerance property is Conjecture 2, never proven. We describe a newest axis, zkinfer (zkML proof-of-inference), which is MODELED from five peer-reviewed sources rather than run at production scale. The paper's centerpiece is an explicit statement of what we do not claim: receipts attest provenance, not output correctness; there is no production Authority-to-Operate; and supply-chain hardening is honestly reported at SLSA L1 (attested toward L2, with L3 on the roadmap). All claims are reproducible against live endpoints anyone can curl.

Keywords: verifiable AI · governed inference · signed receipts · formal methods · Lean 4 · zero-knowledge machine learning · trust bounds · honest-by-design. Suggested arXiv: cs.CR (primary); cs.LG, cs.SE (cross-list).

1 The honest hook

Most “verifiable AI” and “trustworthy AI” launches win attention by overclaiming: they imply formal guarantees they do not hold, or present advisory scores as proofs. The projects that actually earned durable credibility did the opposite — they shipped a small, checkable cryptographic primitive and then stated their own boundaries out loud. NotaryOS framed its contribution as “cryptographic proof of what your AI agent chose not to do,” and won trust by openly posting its hardest open limitation¹³. Circe shipped a one-command verifier that passes on a clean receipt and fails on a single mutated field, and explicitly disclaimed that it is “not a logging system, observability platform, or policy engine”¹⁴. DRM3 grounds its “no trust required” tagline in a published signer registry and real status counts rather than superlatives¹⁵.

The SZL estate takes the same posture and adds one thing none of those have: a live, clickable, 63-surface estate already running behind the primitive^{3,2}. Our hook is therefore not a promise of trustworthiness — it is an invitation to audit:

THE ONE-SENTENCE HOOK

A live, 63-surface governed-AI estate where every inference emits a signed receipt, a handful of core invariants are machine-checked in Lean, and the advisory trust score is honestly capped below 1.0 — because “100% trustworthy AI” is a claim no honest system can make.

2 System overview

The estate is organized as a hub of independent holographic surfaces, each a self-contained 3D visualization backed by a live API route. The public surfaces API returns “count”: 63 and enumerates every surface with a verbatim honesty label parsed from its source (never upgraded)². Anyone can confirm the count and inspect the estate directly:

² Live surfaces API (returns “count”: 63): <https://a-11-oy.com/api/a11oy/v1/frontier/surfaces>

³ SZL Holographic Estate (63 live surfaces): <https://a-11-oy.com/holographic>

⁴ Living-Anatomy vitals API (MEASURED joules when a GPU exporter is reachable, else UNAVAILABLE): <https://a-11-oy.com/api/a11oy/v1/anatomy/vitals>

¹³ NotaryOS — Show HN (Ed25519 hash-chained receipts; counterfactual receipts): <https://news.ycombinator.com/item?id=47193330>

¹⁴ Circe — Show HN (offline-verifiable receipts over RFC 8785 canonical JSON): <https://news.ycombinator.com/item?id=46687551>

¹⁵ DRM3 Provenance SDK — attested provenance, published signer registry: <https://drm3.io/products/provenance>

```
$ curl -s https://a-11-oy.com/api/a11oy/v1/frontier/surfaces | jq '.count'
63

$ curl -s https://a-11-oy.com/api/a11oy/v1/anatomy/vitals | jq '.meta.lungs.label'
"MEASURED" # when a GPU exporter is reachable; otherwise "UNAVAILABLE" — never fabricated
```

2.1 Living-anatomy telemetry (MEASURED joules)

A living-anatomy layer projects the estate as body systems over real telemetry. GPU energy is reported in joules and is labelled MEASURED only when read from a reachable NVML-style exporter; when no exporter is reachable, the field is honestly reported as UNAVAILABLE rather than filled with a modelled value⁴. The projection is explicitly framed as a view — SZL is never claimed to be literally alive. This is the estate's honesty discipline made mechanical: a field is either measured, modelled, or unavailable, and it says which.

2.2 Governed-inference receipts

Each governed inference emits a signed receipt. The receipt is a canonical-JSON object¹⁷ that binds the identity of the model artifact, the configuration, and the content hash of the run, signed with an Ed25519 key. It is tamper-evident and offline-verifiable: a single mutated field breaks the signature. This mirrors the primitive currently earning traction in the field^{13,14} and aligns with the emerging IETF verifiable-inference-chain conventions¹⁶. What the receipt attests is precisely provenance of the inference — what ran, with which config, and its hash — and nothing more (see §6).

2.3 Existing social proof: the killinchu OSINT corpus

The estate's one substantial external traction magnet is the killinchu-osint-corpus on Hugging Face, with 15,576 downloads in the last month at retrieval time⁵. It anchors the estate's public presence and serves as the funnel toward the live surfaces.

3 The formal core: the locked-8, Λ , and BFT

The estate's formal guarantees are deliberately narrow and precisely bounded. Exactly eight invariants — the locked-8 — are machine-checked in Lean 4 at kernel c7c0ba17:

F1	F4	F7	F11	F12	F18	F19	F22
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The locked-8, machine-checked in Lean 4 at kernel c7c0ba17. New surfaces add nothing to this set.

These eight, and only these eight, are proven. New surfaces — including the newest axis in §5 — add nothing to the locked-8 and touch neither a locked formula nor the kernel¹. The discipline of publishing the boundary — exactly what is proven and what is not — is the estate's core credibility move.

3.1 Λ is Conjecture 1 — explicitly not a theorem

STATUS OF Λ (LAMBDA)

Λ , the estate's advisory trust functional, is Conjecture 1. It is advisory only and is never presented as a theorem, never rendered green, and never assigned the value 1.0. Its associated trust score is capped at 0.97 and never reaches 100%. Treating Λ as proven would be a category error the estate refuses to make.

The 0.97 ceiling is a design decision, not a measured accuracy: it encodes the position that no operating AI system can honestly claim a trust value of 1.0. A score that cannot reach 1.0 is a feature — it keeps the advisory nature of Λ legible at a glance.

⁴ Living-Anatomy vitals API (MEASURED joules when a GPU exporter is reachable, else UNAVAILABLE): <https://a-11-oy.com/api/a11oy/v1/anatomy/vitals>

¹⁷ RFC 8785, JSON Canonicalization Scheme (JCS): <https://www.rfc-editor.org/rfc/rfc8785>

¹³ NotaryOS — Show HN (Ed25519 hash-chained receipts; counterfactual receipts): <https://news.ycombinator.com/item?id=47193330>

¹⁴ Circe — Show HN (offline-verifiable receipts over RFC 8785 canonical JSON): <https://news.ycombinator.com/item?id=46687551>

¹⁶ IETF I-D, "Cryptographically Verifiable Inference Chain for AI Agents": <https://www.ietf.org/archive/id/draft-mw-spice-inference-chain-00.html>

⁵ killinchu-osint-corpus, Hugging Face (15,576 downloads last month, retrieved 2026-07-06): <https://huggingface.co/datasets/SZLHOLDINGS/killinchu-osint-corpus>

¹ SZL Holdings internal charter & live estate audit (doctrine v1), 2026-07-06. Verified live endpoint: <https://a-11-oy.com/api/a11oy/v1/frontier/surfaces>

3.2 BFT is Conjecture 2

The estate's Byzantine-fault-tolerance property is Conjecture 2 and is likewise never proven. Like Λ , it is labelled as a conjecture wherever it appears; it is not part of the locked-8 and carries no theorem status¹.

Element	Status	What it means
Locked-8 {F1,F4,F7,F11,F12,F18,F19,F22}	PROVEN	Machine-checked in Lean 4 at kernel c7c0ba17. Exactly eight — never "8+".
Λ (trust functional)	CONJECTURE 1	Advisory only; never a theorem; score capped at 0.97, never 1.0.
BFT	CONJECTURE 2	Never proven; labelled conjecture wherever shown.
Governed-inference receipt	ATTESTED	Signed, offline-verifiable provenance of a run — not output correctness.
zkinfer (new axis)	MODELED	Parametric cost model from literature; not a live LLM-scale proof.

Table 1. Honest status of each formal element. The boundary — what is proven vs. conjectured vs. modelled — is published deliberately.

4 The newest axis: zkinfer (zkML proof-of-inference)

The estate's newest axis, zkinfer — "zkML Proof-of-Inference (Cryptographic Receipts)" — is being built as an additive surface. Its honesty label is MODELED. It is the cryptographic-proof branch of verifiable inference: a prover emits a succinct zero-knowledge argument, checkable by anyone holding only a public commitment to the weights, that a specific committed model produced a specific output on a given input — without revealing the parameters and without the verifier re-running the model. This is orthogonal to the estate's existing hardware/TEE attestation branch (ccattest): zkML places no trusted hardware and no silicon vendor in the trust base, relying only on standard cryptographic hardness assumptions.

4.1 What is MODELED, and what is not

HONESTY LABEL: MODELED (not VERIFIED)

The estate is not running a production zk-SNARK prover over a live LLM forward pass. The zkinfer endpoint returns modelled, parametric quantities (prover time, verify time, proof size as functions of model size, sequence length, and proof system) derived from the five peer-reviewed sources below, each returned value carrying its citing arXiv ID / DOI. It never claims a green/VERIFIED state and never claims a live proof was produced unless one actually was. Where a small, genuine commit-and-verify roundtrip can be run honestly at request time it is labelled narrowly as MEASURED for that claim only; otherwise it downgrades to HONEST-STUB — a passing proof is never fabricated.

4.2 Basis in the literature (five verified sources)

The cost model is parameterized from established zkML results: zkLLM proves full-inference correctness for a 13B-parameter model in under 15 minutes with a proof under 200 kB while hiding parameters, via its tlookup and zkAttn constructions⁶; Kang et al. give the first practical ImageNet-scale non-interactive ZK-SNARK proof of valid inference and the MLaaS verification protocols⁷; the ZKML (EuroSys 2024) system is an optimizing compiler from TensorFlow to halo2 circuits, reporting up to 5× larger provable models, 5× faster verification, and 22× smaller proofs versus prior work⁸; South et al. package ZK inference proofs into verifiable evaluation attestations — a model with fixed private weights provably achieving a stated benchmark, which maps directly onto a receipt⁹; and the 2025 survey by Peng et al. consolidates the field and documents the three honest bottlenecks the surface must show — limited circuit expressiveness, high proving cost, and deployment complexity¹⁰.

¹ SZL Holdings internal charter & live estate audit (doctrine v11), 2026-07-06. Verified live endpoint: <https://a-11-oy.com/api/a11oy/v1/frontier/surfaces>

⁶ Sun, Li, Zhang, "zkLLM: Zero Knowledge Proofs for Large Language Models," ACM CCS 2024, arXiv:2404.16109: <https://arxiv.org/abs/2404.16109>

⁷ Kang, Hashimoto, Stoica, Sun, "Scaling up Trustless DNN Inference with Zero-Knowledge Proofs," 2022, arXiv:2210.08674: <https://arxiv.org/abs/2210.08674>

⁸ Chen et al., "ZKML: An Optimizing System for ML Inference in Zero-Knowledge Proofs," EuroSys 2024, DOI 10.1145/3627703.3650088: <https://doi.org/10.1145/3627703.3650088>

⁹ South, Camuto, Jain et al., "Verifiable evaluations of machine learning models using zkSNARKs," 2024, arXiv:2402.02675: <https://arxiv.org/abs/2402.02675>

¹⁰ Peng, Wang, Zhao et al., "A Survey of Zero-Knowledge Proof Based Verifiable Machine Learning," 2025, arXiv:2502.18535: <https://arxiv.org/abs/2502.18535>

Because these numbers come from the cited literature and are labelled MODELED, they are defensible without any claim that the estate reproduced them at scale. The surface's value is that it puts the cryptographic branch beside the estate's existing hardware branch and shows the trust-model contrast explicitly.

5 Limitations & what we do NOT claim

This section is the credibility centerpiece. The estate's advantage is not that it claims more than comparable work — it is that it states its boundaries plainly.

1. Receipts attest provenance, not correctness.

A signed receipt proves what ran, with which configuration, and its hash — tamper-evident provenance of the inference. It does not prove that the output is correct, safe, unbiased, or fit for any purpose. We say “tamper-evident provenance,” never “proof of correctness.”

2. Λ is a conjecture; trust never reaches 1.0.

Λ is Conjecture 1 and BFT is Conjecture 2. Neither is a theorem. The advisory trust score is capped at 0.97 and is never 100%. We do not sell “trustworthy AI.”

3. Exactly eight invariants are proven — not more.

The locked-8 is {F1,F4,F7,F11,F12,F18,F19,F22} at kernel c7c0ba17. We never write “8+,” never add surfaces to the locked set, and never call Λ proven.

4. zkinfer is MODELED, not a live LLM-scale proof.

The zkML cost figures are parametric estimates from the literature. No production zk-SNARK prover runs over a live LLM forward pass; the label is MODELED, and any real micro-artifact is scoped narrowly (MEASURED or HONEST-STUB).

5. No production Authority-to-Operate (ATO).

The estate is a live research/demonstration system. It carries no production ATO, no accreditation, and no warranty of fitness for regulated or mission use.

6. Energy is MEASURED or UNAVAILABLE — never modelled-as-measured.

GPU joules are reported as MEASURED only from a reachable exporter; otherwise the field reads UNAVAILABLE. We never present a modelled energy figure as if it were measured.

5.1 Supply-chain hardening: honest SLSA posture

SLSA level	Estate status	Meaning
L1	HONEST (current)	Build process exists and provenance is generated/available.
L2	ATTESTED (in progress)	Signed provenance from a hosted build platform; tamper-resistant against the build itself.
L3	ROADMAP	Hardened, non-falsifiable provenance with strong isolation. Not yet claimed.

We report SLSA L1 as honest current state, L2 as attested / in progress, and L3 explicitly as roadmap¹⁸ — never claiming a level the estate has not reached.

¹⁸ SLSA v1.0 — Supply-chain Levels for Software Artifacts (build levels L1–L3): <https://slsa.dev/spec/v1.0/levels>

6 Reproducibility

Every headline claim in this paper is checkable against a live endpoint. Reviewers do not have to take our word for it — they can run the commands below and inspect the estate in a browser^{3,2,4,5}.

```
# 63 live surfaces (expect: 63)
curl -s https://a-11-oy.com/api/a11oy/v1/frontier/surfaces | jq '.count'

# per-surface honesty labels (verbatim, never upgraded)
curl -s https://a-11-oy.com/api/a11oy/v1/frontier/surfaces | jq '.surfaces[] | {id,label}'

# living-anatomy vitals: MEASURED joules when an exporter is reachable, else UNAVAILABLE
curl -s https://a-11-oy.com/api/a11oy/v1/anatomy/vitals | jq '.metal.lungs.label'

# open the estate in a browser
open https://a-11-oy.com/holographic
```

For citability we intend to deposit this whitepaper and its artifacts (the receipt verifier, the Lean proof kernel c7c0ba17, and a corpus snapshot) to Zenodo under CC-BY-4.0 for a version DOI and a concept DOI, cross-linked to the arXiv identifier and the Hugging Face org. The arXiv submission targets cs.CR (primary) with cs.LG and cs.SE cross-lists.

7 Related work & comparables

The estate sits alongside a growing body of verifiable-AI work. On the systems-and-thesis side, South's 2025 thesis gives practical blueprints combining zero-knowledge cryptography, TEEs, and MPC for verifiable and auditable claims about foundation models¹¹. Cankaya reframes GPU non-determinism as an auditable fingerprint via bit-exact re-computation of inference¹² — directly relevant to a deterministic-receipt story. On the primitive side, NotaryOS¹³, Circe¹⁴, and the DRM3 Provenance SDK¹⁵ demonstrate the exact traction pattern the estate follows: a small verifiable primitive plus a candid statement of limits. The IETF verifiable-inference-chain draft¹⁶ and RFC 8785 canonical JSON¹⁷ indicate the space is standardizing, and aligning the receipt schema to those conventions buys interoperability credibility. What distinguishes the SZL estate is that the primitive already sits behind a live, 63-surface, publicly inspectable system.

Author. Stephen P. Lutar Jr., SZL Holdings. Correspondence & live estate: <https://a-11-oy.com/holographic>. Corpus: <https://huggingface.co/datasets/SZLHOLDINGS/killinchu-osint-corpus>.

Honest by design (doctrine v11). No metric, DOI, or proof in this document is fabricated; $\wedge$ is a conjecture, not a theorem; trust is capped at 0.97 and never 100%; the estate serves 63 live surfaces.

³ SZL Holographic Estate (63 live surfaces): <https://a-11-oy.com/holographic>

² Live surfaces API (returns "count": 63): <https://a-11-oy.com/api/a11oy/v1/frontier/surfaces>

⁴ Living-Anatomy vitals API (MEASURED joules when a GPU exporter is reachable, else UNAVAILABLE): <https://a-11-oy.com/api/a11oy/v1/anatomy/vitals>

⁵ killinchu-osint-corpus, Hugging Face (15,576 downloads last month, retrieved 2026-07-06): <https://huggingface.co/datasets/SZLHOLDINGS/killinchu-osint-corpus>

¹¹ South, "Private, Verifiable, and Auditable AI Systems," 2025, arXiv:2509.00085: <https://arxiv.org/abs/2509.00085>

¹² Cankaya, "Bit-Exact AI Inference Verification Without Performance Tradeoffs," 2026, arXiv:2606.00279: <https://papers.cool/arxiv/2606.00279>

¹³ NotaryOS — Show HN (Ed25519 hash-chained receipts; counterfactual receipts): <https://news.ycombinator.com/item?id=47193330>

¹⁴ Circe — Show HN (offline-verifiable receipts over RFC 8785 canonical JSON): <https://news.ycombinator.com/item?id=46687551>

¹⁵ DRM3 Provenance SDK — attested provenance, published signer registry: <https://drm3.io/products/provenance>

¹⁶ IETF I-D, "Cryptographically Verifiable Inference Chain for AI Agents": <https://www.ietf.org/archive/id/draft-mw-spice-inference-chain-00.html>

¹⁷ RFC 8785, JSON Canonicalization Scheme (JCS): <https://www.rfc-editor.org/rfc/rfc8785>

¹ SZL Holdings internal charter & live estate audit (doctrine v11), 2026-07-06. Verified live endpoint: <https://a-11-oy.com/api/a11oy/v1/frontier/surfaces>